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Kelvin Probe Force Microscopy and Raman Spectroscopy of Graphene Grown on 1095 Steel

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Kelvin probe force microscopy (KPFM) is an atomic force microscope (AFM) technique well suited for evaluating graphene as an anticorrosive coating for steel, because of the close relationship between electrical work function and electrochemical corrosion potential [1]. In KPFM, a compensating bias—referred to as the contact potential difference (V_{CPD})—is applied between a metal-coated probe and the sample to nullify electrostatic forces. This allows spatial mapping of relative work functions across the surface [2]. Graphene's chemical inertness, impermeability, and tunable electronic properties make it a promising corrosion barrier, with work function strongly influenced by layer number, defect density and interaction with the underlying substrate.

Here, graphene was grown directly on 1095 steel using chemical vapor deposition (CVD) at 600 °C using acetylene and hydrogen. Characterization used Raman spectroscopy, AFM/KPFM in tapping mode, and optical microscopy. Our group previously showed that CVD conditions, alloy composition, and microstructures influence graphene layer number, and defect density, on steel substrates [3].

Fig. 1a shows the limitations of optical microscopy for characterizing graphene grown on steel; the graphene is indiscernible due to contrast from surface structures. Likewise, AFM topography images in Fig. 1b are limited by the scale of surface structures compared to graphene's thickness. With KPFM, Fig. 1c, contrast is decoupled from topography, relying on work function relative to the AFM probe. This enables identification of nanoscale to microscale graphene areas on the rough substrate, and differentiation between graphene regions with varying layer number, defects, and coverage based on the local V_{CPD} .

Areas labeled as single-layer and few-layer graphene (SLG, FLG) have the lowest V_{CPD} , indicating a higher relative work function, while multilayer graphene (MLG) shows intermediate V_{CPD} . A reasonable hypothesis is that SLG is strongly chemisorbed to the steel, doping the graphene and increasing the work function, while additional graphene layers in MLG screen this effect. Areas with low coverage have the highest V_{CPD} , due to exposed steel and defects like graphene edges. These differences in V_{CPD} show the importance of high coverage and low defects for anticorrosive graphene coatings.

Raman point spectra in Fig. 2a correspond to labeled spots in Fig. 1c, supporting the KPFM interpretation. The red spectrum from the high V_{CPD} area has a high D:G peak ratio, consistent with more graphene defects and low coverage. In contrast, the MLG (green), FLG (blue) and SLG (black) have low D:G peak ratios indicative of a coherent, low-defect coating. FLG and SLG were identified by narrow 2D peaks, with a 2D:G peak ratio > 1 as the criterion for SLG. The G peaks of FLG and SLG are also blue-shifted, supporting the chemisorption-induced doping hypothesis [4]. The Raman maps of the D:G ratio in Fig. 2b and 2D:G ratio in Fig. 2c show strong spatial correlation between low defect density, thinner graphene and low V_{CPD} .

KPFM effectively reveals graphene grown on steel, and the importance of layer number, defects, and coverage on work function. While high-coverage, low-defect MLG increases the work function, stronger interactions between FLG or SLG and the steel surface raise the work function even more. Combined with Raman spectroscopy, KPFM helps identify suitable CVD parameters that promote high coverage, low defect density, and controlled layer numbers for graphene anticorrosive coatings on steel.

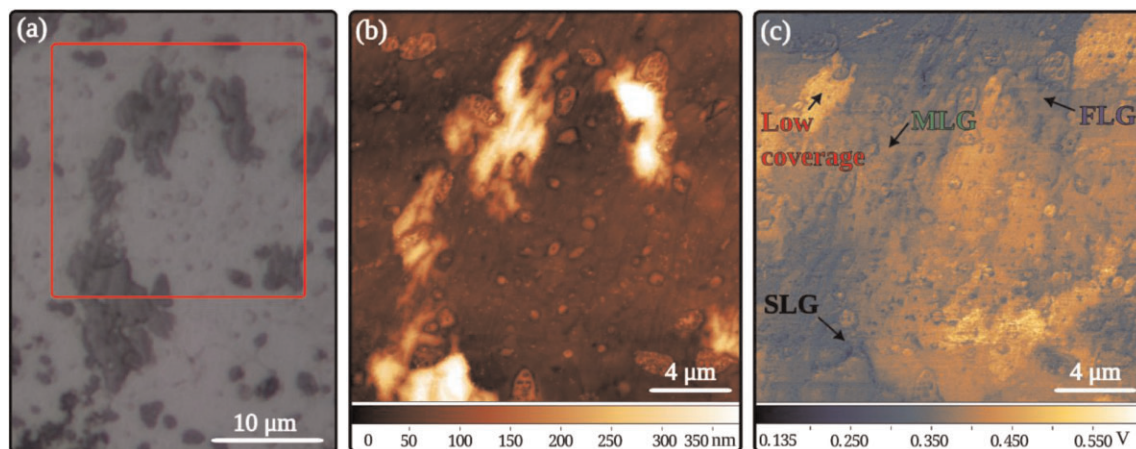


Fig. 1. (a) Optical image of graphene coating. The red box shows the region imaged in (b) and (c). (b) AFM topography image of rough steel surface. (c) KPFM image revealing effects of graphene on V_{CPD} . Areas with SLG and FLG have the lowest V_{CPD} (highest work function), areas with MLG have intermediate V_{CPD} , and low coverage areas have high V_{CPD} (most susceptible to corrosion).

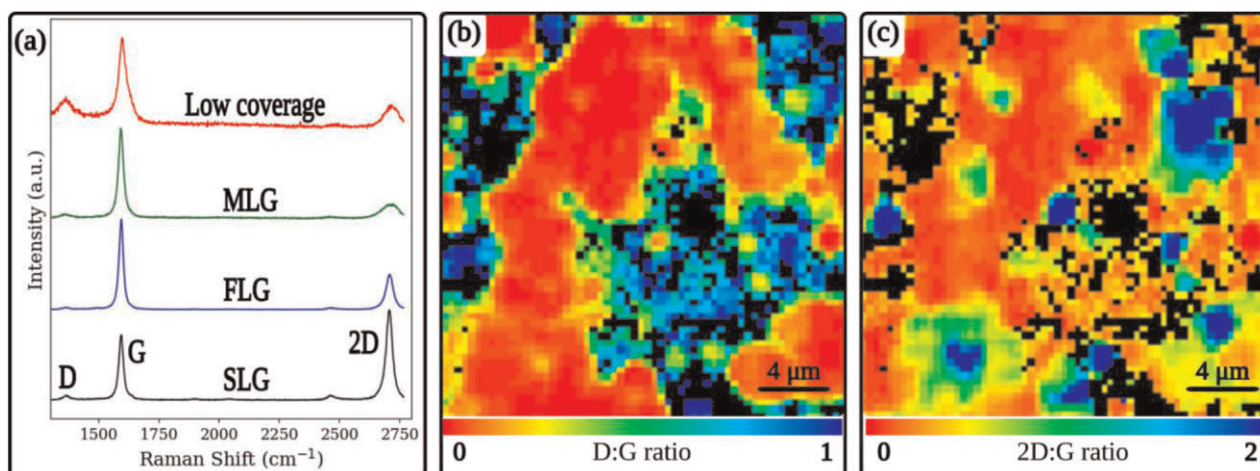


Fig. 2. (a) Representative Raman spectra collected at the spots labeled in Fig. 1c. (b) D:G peak ratio map showing that a low D-peak corresponds to low V_{CPD} areas. Black pixels indicate spectra with peak intensities below 10 standard deviations of the background due to low coverage. (c) 2D:G peak ratio map where the highest ratio typically corresponds to lowest V_{CPD} .

References

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